Couette flow

Wiki *Enhanced dissipation for passive scalars*

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Discuss on Zulip: #examples, topic "Couette flow".

The Couette flow is the linear shear $u = (y, 0)$, considered here in the simplest possible domain for this problem, $(x, y) \in \mathbb{T} \times \mathbb{R}$. The equation under study then reduces to

$$\partial_t f + y\partial_x f = \nu \Delta f, \qquad f|_{t=0} = f^{\mathrm{in}}, \qquad (x, y) \in \mathbb{T} \times \mathbb{R}.$$

This is a foundational model in the field, because the same equation governs the vorticity of the two-dimensional Navier–Stokes equations linearized around the Couette flow. In that context Kelvin already solved it explicitly in 1887 (see Kelvin (1887, eq. (34)–(35))). Moreover, the same model, without the diffusion in x, was derived by Kolmogorov (1934) when developing his theory of Brownian motion, and the equation is also of Fokker–Planck type. See also History of the problem.

We use this model to illustrate what typical mixing and enhanced dissipation results look like, and how the two are related. The model is simple enough that each proof technique can be carried out in full. They are all presented below on this single example.

Why this domain? The domain $\mathbb{T} \times \mathbb{R}$ is mathematically convenient: there are no boundaries, and one can take the Fourier transform in both variables.
There is also a more geometric motivation. Consider a circular flow in the plane, $u = rV(r)e_\theta$, for which the equation reads $\partial_t f + V(r)\partial_\theta f = \nu \Delta f$. The log-polar coordinates $X = e^s \cos(\theta)$, $Y = e^s \sin(\theta)$ identify $\mathbb{R}^2 \setminus \{0\}$ with $(\theta, s) \in \mathbb{T} \times \mathbb{R}$, as shown in the figure below.

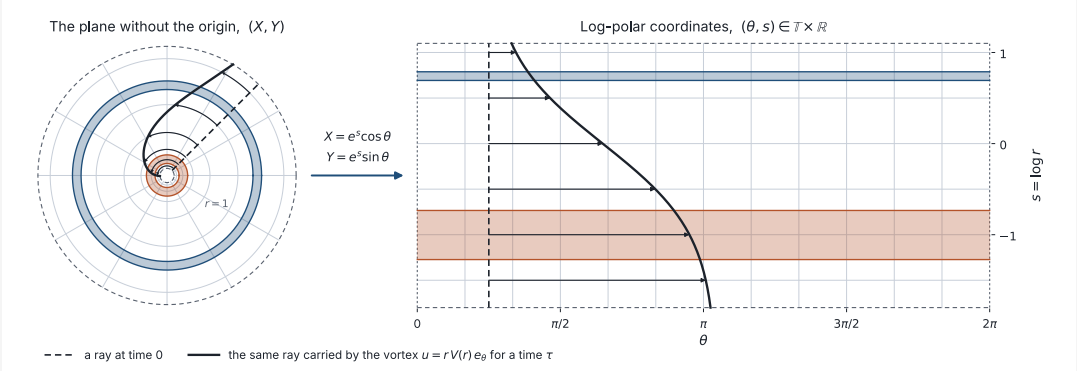


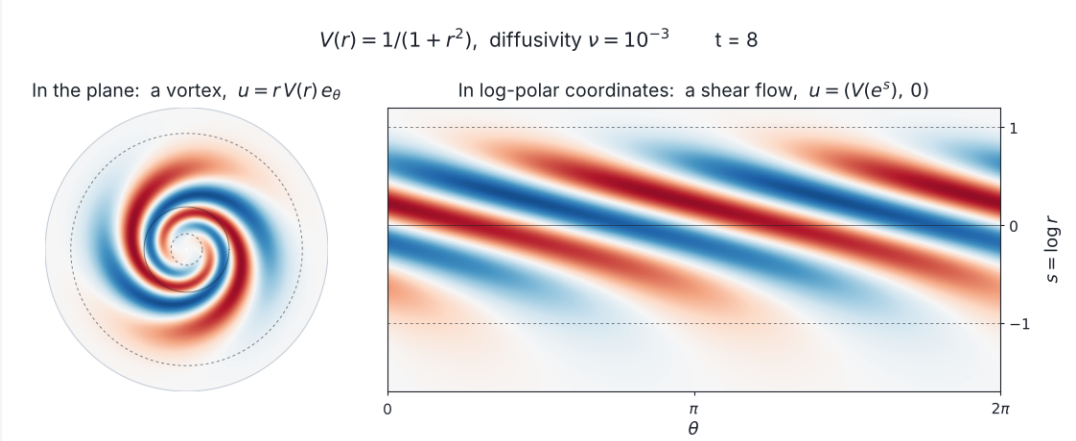
Figure. Left: the plane, with two annuli of the same width and a ray (dashed) carried by a circular flow for some time (solid). Right: the same picture in log-polar coordinates. The annuli become strips, and the circular flow becomes a shear flow.

Note that this is a conformal change of variables, so the Laplacian only gains a scalar

factor, and the equation becomes

$$\partial_t f + V(e^s)\partial_\theta f = \nu e^{-2s}(\partial_\theta^2 + \partial_s^2) f, \qquad (\theta, s) \in \mathbb{T} \times \mathbb{R}.$$

If we localize near a radius $r_0 = e^{s_0}$ where $V'(r_0) \neq 0$, then to leading order, and up to constants, we recover the equation for the Couette flow. In this sense, the Couette flow can be seen as a first local approximation of a radial vortex in $\mathbb{R}^2$, in a region where the angular velocity is strictly monotone. Below is a video showing the effect of the conformal mapping.



Video. Left: a scalar transported and diffused by the circular flow with $V(r) = 1/(1 + r^2)$. Right: the same solution in log-polar coordinates, where the flow is a shear flow.
The video plays in the online edition of this page. How it was made: page Numerical methods, record `num-log-polar`.

1 The result

Before introducing the main theorem, it is useful to observe that, taking the x-average, one is left with the standard heat equation on $\mathbb{R}$. Namely,

$$\langle f \rangle_x = \frac{1}{2\pi} \int_{\mathbb{T}} f(x, y)\, dx \quad \Longrightarrow \quad \langle f(t) \rangle_x = e^{\nu t \partial_{yy}} \langle f^{\mathrm{in}} \rangle_x.$$

For this part of the solution there is therefore nothing new to say, and it enjoys the standard diffusive estimates. Hence, the quantity to measure is how fast the solution approaches its x-average.

Theorem 1. *Let $\nu \in [0, 1)$ and let f be the solution with initial datum $f^{\mathrm{in}} \in L^2(\mathbb{T} \times \mathbb{R})$. There exist constants $c, C > 0$, independent of ν, such that for all $t \geq 0$ the following inequality holds true:*

$$\|(f - \langle f \rangle_x)(t)\|_{L^2} \leq C e^{-c\nu^{1/3} t} \|f^{\mathrm{in}} - \langle f^{\mathrm{in}} \rangle_x\|_{L^2}.$$

Moreover, if $f^{\mathrm{in}} \in H^1(\mathbb{T} \times \mathbb{R})$, the following mixing estimate, uniform in ν, holds true:

$$\|(f - \langle f \rangle_x)(t)\|_{H^{-1}} \leq \frac{C}{1 + t} e^{-c\nu^{1/3} t} \|f^{\mathrm{in}} - \langle f^{\mathrm{in}} \rangle_x\|_{H^1}.$$

The x-average as a projection. The x-average is in fact nothing more than the orthogonal projection in L^2 onto the kernel of the transport operator $y\partial_x$: the functions in the kernel

are exactly those that do not depend on x. When the transport operator has a non-trivial kernel, we cannot expect the elements of the kernel to experience any mixing at all from the transport. The interaction with the dissipation is however much more subtle, and it relies heavily on the commutation properties of this projection with the diffusion operator. In this very simple case everything trivializes: the projection commutes with $\nu\Delta$, so the x-average solves the heat equation and $f - \langle f \rangle_x$ solves the same equation as f. But the mathematical structure behind the convergence towards the x-average lies in the fact that the transport operator has a non-trivial kernel.

Why negative Sobolev norms measure mixing. The L^2 norm does not see mixing: for $\nu = 0$ it is conserved, whatever the flow does. The norm of H^{-1} does, because it measures how much of a function has been sent to high frequencies: in Fourier, it weighs each mode by the inverse of the size of its frequency, so it is small when the function lives at high frequencies, even if its L^2 norm is not. On the physical side, negative Sobolev norms scale as powers of a length: the ratio $\|g\|_{H^{-1}}/\|g\|_{L^2}$ can be thought of as the length at which g oscillates. For $\nu = 0$, the theorem says that this length decays like $1/t$. See also the note on mixing in the Overview.

Other function spaces. The theorem is stated in spaces based on L^2. Since the flow is incompressible, the equation does not increase any L^p norm. For $1 < p < \infty$, enhanced dissipation in L^p then follows from the L^2 estimate by interpolation, with a rate that degenerates as $p \to 1$ and as $p \to \infty$. Some techniques give the estimate in L^p directly, without this loss. For the Couette flow, the explicit solution gives the first estimate of the theorem with L^p in place of L^2 for every $p \in [1, \infty]$, with constants independent of p.

Idea. The key idea underlying all the methods below is that the transport operator $y\partial_x$ does not commute with the diffusion $\nu\Delta$: indeed, $[\Delta, y\partial_x] = 2\partial_x\partial_y$. If the two commuted, the solution would be $e^{-ty\partial_x}e^{\nu t\Delta}f^{\text{in}}$ and, since the transport preserves the L^2 norm, the flow would have no effect on the decay. We will highlight in each method where this lack of commutation enters.

At first sight, operators that do not commute look like an annoying technical problem. This is a typical instance where a seemingly unfavorable property, looked at in the right way, turns out to be exactly what helps.

2 Notation and setting

Throughout, $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$. For the Fourier transform in x we write

$$f(t,x,y) = \sum_{k\in\mathbb{Z}} f_k(t,y)e^{ikx}, \qquad f_k(t,y) = \frac{1}{2\pi}\int_{\mathbb{T}} f(t,x,y)e^{-ikx}\,dx,$$

so that $f_0 = \langle f \rangle_x$ and $\|f\|_{L^2}^2 = 2\pi\sum_k \|f_k\|_{L^2(\mathbb{R})}^2$. Note that in this model each k-mode evolves independently of the others:

$$\partial_t f_k + iky f_k = \nu(\partial_{yy} - k^2)f_k, \qquad k \in \mathbb{Z}.$$

The Sobolev norms in the theorem are $\|g\|_{H^1}^2 = \|g\|_{L^2}^2 + \|\nabla g\|_{L^2}^2$ and its dual,

$$\|g\|_{H^{-1}} = \sup\Big\{ \int_{\mathbb{T}\times\mathbb{R}} g\varphi\,dx\,dy \;:\; \|\varphi\|_{H^1} \le 1\Big\}.$$

3 One example, every technique

Each proof below is written for this example only, to show the mechanism; the general results, with complete proofs, are in the part *Techniques and general results*.

3.1 Explicit solution

Proof by the explicit solution. [The proof.] □

3.2 Spectral methods

Proof by spectral methods. [The proof.] □

General theory: Spectral methods.

3.3 Hypocoercivity

Proof by hypocoercivity. [The proof.] □

General theory: Hypocoercivity.

3.4 Stochastic methods

Proof by stochastic methods. [The proof.] □

General theory: Stochastic methods.

3.5 Hörmander theory

Proof by Hörmander theory. [The proof.] □

General theory: Hörmander theory.

4 What this example teaches

[What the example shows, and what the general theory keeps of it.]

References

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W. Thomson (Lord Kelvin), *Stability of fluid motion: rectilineal motion of viscous fluid between two parallel planes*, Philosophical Magazine (5) **24** (1887), 188–196, doi:10.1080/14786448708628078.